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Microsoft Windows [Version 10.0.26200.7623]
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C:\Users\Lenovo-User>cd desktop

C:\Users\Lenovo-User\Desktop>md expense_predictor_api

C:\Users\Lenovo-User\Desktop>cd expense_predictor_api

C:\Users\Lenovo-User\Desktop\expense_predictor_api>python -m venv venv

C:\Users\Lenovo-User\Desktop\expense_predictor_api>venv\scripts\activate

(venv) C:\Users\Lenovo-User\Desktop\expense_predictor_api>pip install Flask numpy pandas matplotlib scikit-learn joblib
Collecting Flask
  Using cached flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting numpy
  Downloading numpy-2.4.3-cp314-cp314-win_amd64.whl.metadata (6.6 kB)
Collecting pandas
  Downloading pandas-3.0.1-cp314-cp314-win_amd64.whl.metadata (19 kB)
Collecting matplotlib
  Downloading matplotlib-3.10.8-cp314-cp314-win_amd64.whl.metadata (52 kB)
Collecting scikit-learn
  Downloading scikit-learn-1.8.0-cp314-cp314-win_amd64.whl.metadata (11 kB)
Collecting joblib
  Downloading joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)
Collecting blinker>=1.9.0 (from Flask)
  Using cached blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from Flask)
  Using cached click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
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Downloading fonttools-4.62.1-cp314-cp314-win_amd64.whl (2.3 MB)
  2.3/2.3 MB 2.6 MB/s 0:00:00
Using cached itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading kiwisolver-1.5.0-cp314-cp314-win_amd64.whl (75 kB)
Using cached markupsafe-3.0.3-cp314-cp314-win_amd64.whl (15 kB)
Downloading packaging-26.0-py3-none-any.whl (74 kB)
Downloading pillow-12.1.1-cp314-cp314-win_amd64.whl (7.2 MB)
  7.2/7.2 MB 2.8 MB/s 0:00:02
Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading scipy-1.17.1-cp314-cp314-win_amd64.whl (37.3 MB)
  37.3/37.3 MB 3.3 MB/s 0:00:11
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Using cached werkzeug-3.1.6-py3-none-any.whl (225 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading tzdata-2025.3-py2.py3-none-any.whl (348 kB)
Installing collected packages: tzdata, threadpoolctl, six, pyparsing, pillow, packaging, numpy, markupsafe, kiwisolver,
joblib, itsdangerous, fonttools, cycler, colorama, blinker, werkzeug, scipy, python-dateutil, jinja2, contourpy, click,
scikit-learn, pandas, matplotlib, Flask
Successfully installed Flask-3.1.3 blinker-1.9.0 click-8.3.1 colorama-0.4.6 contourpy-1.3.3 cycler-0.12.1 fonttools-4.62
.1 itsdangerous-2.2.0 jinja2-3.1.6 joblib-1.5.3 kiwisolver-1.5.0 markupsafe-3.0.3 matplotlib-3.10.8 numpy-2.4.3 packagin
g-26.0 pandas-3.0.1 pillow-12.1.1 pyparsing-3.3.2 python-dateutil-2.9.0.post0 scikit-learn-1.8.0 scipy-1.17.1 six-1.17.0
threadpoolctl-3.6.0 tzdata-2025.3 werkzeug-3.1.6

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

(venv) C:\Users\Lenovo-User\Desktop\expense_predictor_api>
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C:\Users\Lenovo-User\Desktop\expense_predictor_api\generate_data.py • - Sublime Te...
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app.py generate_data.py home.html x
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 # ---Config for data generations
6 num_months = 36
7 start_date = '2023-01-01'
8 base_expense = 10000
9 monthly_increase = 200
10 random_flactuation = 1500
11
12 print('--- Generating Sythetic Business Expense Data ---')
13
14 # creating a range of monthly dates
15 dates = pd.date_range(start_date, periods = num_months, freq = 'M')
16
17 # Generate expense
18 expenses = []
19 for i in range(num_months):
20     expense = base_expense + (i*monthly_increase)+ np.random.uniform(-random_flactuation, random_flactuation)
21     expenses.append(max(5000, round(expense,2)))
22
23 #Create a pandas dataframe
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C:\Users\Lenovo-User\Desktop\expense_predictor_api\train_model.py - Sublime Text (...
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app.py — generate_data.py x app.py — expense_predictor_api x train_model.py x

1 import pandas as pd
2 import numpy as np
3 from sklearn.linear_model import LinearRegression
4 from sklearn.model_selection import train_test_split
5 from sklearn.metrics import mean_absolute_error, r2_score
6 import joblib
7 import matplotlib.pyplot as plt
8 from matplotlib import pyplot as plt
9
10 print('--- Training Smart Business Expense Predictor Model ---')
11
12 try:
13     data = pd.read_csv('monthly_expenses.csv', index_col = 'Month')
14     print('Data loaded successfully. Shape:', data.shape)
15 except FileNotFoundError:
16     print('Error: monthly_expense.csv not found. Please run generate_data.py')
17     exit()
18
19 x = np.array(range(1, len(data)+1)).reshape(-1,1)
20 y = data['Expenses_R']
21
```

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app.py — flask_project x generate_data.py x app.py — expense_predictor_api x train_model.py x home.html x

22
23 x_train, y_train = x, y
24 print(f'Train data size: {len(x_train)} samples.')
25
26 model = LinearRegression()
27
28 model.fit(x_train, y_train)
29
30 print('Model Training Complete')
31 print(f'Model Coefficient (slope): {model.coef_[0]: .2f} R/month')
32 print(f'Model Intercept: {model.intercept_: .2f}')
33
34 y_pred = model.predict(x_train)
35
36 mae = mean_absolute_error(y_train, y_pred)
37 r2 = r2_score(y_train, y_pred)
38 print('Model Evaluation')
39 print(f'Mean Absolute Error: {mae:.2f}')
40 print(f'R-Squared score: {r2:.2f}')
41
42 #Visualise the generated data
43 plt.figure(figsize = (12, 6))
44 plt.plot(data.index, y, Label = 'Actual Expense', marker = 'o', Linestyle = '-', color = 'blue', markersize=4)
45 plt.plot(data.index, y_pred, Label= 'Predicted Expenses (Model)', Linestyle= '--', color = 'red')
46 plt.title('Monthly Business Expense (Actual vs. Predicted)')
47 plt.ylabel('Expenses (R)')
48 plt.xlabel('Month')
49 plt.grid(True)
50 plt.legend()
51 plt.tight_layout()
52 plt.savefig('model_predictions.png')
53 model_filename = 'expense_predictor_model.pkl'
54 joblib.dump(model, model_filename)
55 print('Machine learning model saved!')
56 plt.show()
```

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File "C:\Users\Lenovo-User\Desktop\expense_predictor_api\venv\Lib\site-packages\matplotlib\
update_props
    raise AttributeError(
        errfmt.format(cls=type(self), prop_name=k),
        name=k)
AttributeError: Line2D.set() got an unexpected keyword argument 'Linestyle'

(venv) C:\Users\Lenovo-User\Desktop\expense_predictor_api>python train_model.py
--- Training Smart Business Expense Predictor Model ---
Data loaded successfully. Shape: (36, 1)
Train data size: 36 samples.
Model Training Complete
Model Coefficient (slope): 182.38 R/month
Model Intercept: 10236.99
Model Evaluation
Mean Absolute Error: 862.83
R-Squared score: 0.79
Machine learning model saved!

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